School of Science C Engineering

CAPSTON DESIGN EGR4402

Document: Capstone Project Proposal
Project Title: Sim2Win: A Team-Agnostic, Event-Based Pre-Match Tactical Recommendation System for Football.
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Project Lifecycle Overview

Phase	Description	Key Activities
Phase 1: Project Definition s Proposal	Identify the problem, scope, and goals	Problem identification, literature review, stakeholder/user analysis, SMART goals, project plan
Phase 2: Requirements s Specification	Define detailed functional and non-functional software requirements	Requirement gathering (surveys, interviews), use case development
Phase 3: Design s Architecture	Design system architecture and software components	UML diagrams, database schema design, UI frames, algorithm selection
Phase 4: Implementation s Prototyping	Develop code modules and integrate	Coding, version control, unit testing, module integration
Phase 5: Testing s Validation	Test software for correctness, performance, and usability	Automated testing, benchmarking, security testing, bug fixes
Phase 6: Finalization s Delivery	Document and present final solution	Prepare technical documentation, final report, user manuals, project demo, presentation

1. Abstract/Executive Summary

Football teams make critical tactical decisions before matches, such as selecting formations and defining playstyles, often based on subjective analysis and expert intuition. While data-driven models exist for predicting football match outcomes, most focus solely on result prediction or team ranking and offer limited guidance on tactical decision-making. Furthermore, many existing approaches rely on team identity or historical rankings, restricting their applicability to unseen teams.

This project proposes a **team-agnostic pre-match tactical decision-support system** that recommends formations and tactical playstyles for a team based on the tactical profile of its opponent. Using event-based football data, the system learns statistical associations between team-level tactical behaviors, formations, and match outcomes. Prior to a match, the model simulates multiple tactical configurations for the user's team and estimates the probability of winning, drawing, or losing for each configuration.

The system does not rely on team names, rankings, or historical dominance, allowing it to generalize to teams not present during training. The innovative aspect of this project lies in reframing match prediction as a **tactical recommendation problem**, combining interpretability, generalization, and practical decision support. The resulting framework aims to assist analysts and coaching staff in preparing opponent-specific tactical strategies using explainable, data-driven insights.

2. Introduction s Background

2.1 Problem Statement

Pre-match tactical decisions in football, such as formation selection and playstyle adaptation, significantly influence match outcomes. Existing predictive models primarily focus on forecasting results rather than supporting tactical planning. Additionally, many models depend on team identity or historical rankings, limiting their use for new or less-documented teams. There is a need for a generalizable and interpretable system that supports pre-match tactical decision-making based on opponent characteristics rather than team reputation.

2.2 Project Context

Sports analytics has increasingly leveraged event-based data to quantify team behaviors such as possession, pressing, and attacking intensity. At the same time, machine learning models and rating systems have been used to predict match outcomes. However, these approaches are rarely adapted to recommend tactical configurations. This project builds upon event-based analytics to bridge the gap between predictive modeling and tactical decision support.

2.3 Intended Audience/Beneficiaries

- Football analysts and coaching staff
- Sports analytics researchers
- Data science students
- Developers of football analytics tools

2.4 Methodology

An Agile development methodology will be adopted, allowing iterative refinement of features, models, and evaluation strategies based on intermediate results.

3. Literature Review / Existing Solutions

The application of Machine Learning (ML) to football analytics has evolved significantly, shifting from simple outcome prediction to complex tactical profiling and style analysis. Recent scholarship highlights the efficacy of dimensionality reduction and the distinction between classification and regression tasks in this domain.

Tactical Analysis and Dimensionality Reduction Given the high dimensionality of event-based football data, feature selection is critical for model performance. Baattite [1] successfully demonstrated the use of Exploratory Factor Analysis (EFA) to uncover underlying playing styles by reducing dataset dimensionality. This study utilized Linear Discriminant Analysis (LDA) for classifying these styles and a Principal Component Regressor (PCR) to predict goal differences, validating the link between defined playing styles and team performance. Similarly, literature reviewed in [3] supports the necessity of feature extraction and aggregation to reduce noise before training linear models, noting that geospatial data is increasingly used to calculate the probability of scoring opportunities.

Prediction Methodologies: Classification vs. Regression A systematic review by Moya et al. [2] identifies classification as the predominant method for predicting discrete match outcomes (Win/Draw/Loss). However, they argue that regression models are instrumental for estimating continuous values, such as the number of goals or shots. The review emphasizes that incorporating metrics like historical goal difference, possession, and individual player performance is crucial not just for prediction, but for "optimizing alignment and adjusting game strategies" based on opponent behaviors [2].

Rating Systems and Limitations Statistical rating systems remain a foundational approach for quantifying team strength. Lasek [4] evaluated multiple systems for the Euro 2016 tournament, including Ordinal Regression and Poisson models, which assume goal scoring rates depend on a team’s attacking strength versus an opponent's defensive skill. While effective for general trends, Lasek noted significant limitations in these models' ability to infer context, such as a team's recent qualitative dip in form, leading to probabilistic errors in tournament forecasts [4]. This aligns with older methods like ELO ratings [3], suggesting a gap remains between purely statistical rating systems and the granular, tactical reality of match-play.

4. Innovation s Contribution

The core innovation of this project is a team-agnostic tactical recommendation framework that transforms match prediction into a pre-match decision-support problem. Instead of predicting outcomes directly, the system evaluates multiple tactical configurations and recommends those associated with higher success probabilities.

Feature / Aspect	**Existing Solutions**	**Proposed Approach**
Team identity dependence	High	None
Tactical recommendations	Not provided	Explicit
Interpretability	Low	High
Generalization	Limited	Strong
Data requirements	High	Event-based only

4.1 Expected Impact Contribution

Academic Contribution

- Introduces a new framing of football prediction as tactical recommendation.
- Demonstrates generalizable modeling without team identity.
- Provides an interpretable use of event-based tactical features.

Practical Contribution

- Supports opponent-specific tactical planning.
- Provides actionable pre-match recommendations.
- Can be applied to teams with limited historical data.

5. Project Objectives s Goals

5.1 Primary Objective

To design and evaluate a team-agnostic pre-match tactical decision-support system that recommends formations and playstyles based on opponent tactical profiles.

5.2 Specific Goals (SMART Criteria)

- **Goal 1:** Engineer pre-match tactical profiles from event-based data.
- **Goal 2:** Define interpretable tactical playstyle representations.
- **Goal 3:** Train a probabilistic model linking tactics, formations, and match outcomes.
- **Goal 4:** Implement a simulation-based recommendation mechanism.
- **Goal 5:** Evaluate generalization using unseen teams using **Tactical Agreement Score** which **is** a custom metric to evaluate the model's quality. I will calculate the percentage of matches where the model's *recommended* formation/tactic aligns with the tactic actually used by the winning team. A high agreement score indicates the model has successfully learned the principles of winning football.

6. Methodology / Technical Approach

Software lifecycle: Agile
Tools & technologies: Apis, Python, NumPy, Pandas, Scikit-learn, Pytorch, Matplotlib and Streamlit.

Time-**Series Aggregation:** Implementation of **Rolling Averages (last 5 matches)** for every team. This transforms raw post-match logs into "Pre-Match Tactical Profiles," ensuring the model predicts based only on information available before kickoff (preventing data leakage).

Dimensionality Reduction: Utilization of techniques like PCA or EFA (as cited in Baattite, 2023) to distill complex metrics into interpretable "Playing Style" vectors (e.g., "High Press," "Counterattack").

Architecture:

1. **Data Ingestion Layer:** Aggregation of multi-league match data (EPL, La Liga, etc.).
2. **Preprocessing Layer:** Calculation of Rolling Averages (5-game window) => Normalization.
3. **Modeling Layer:**
   Input: Team Style Vector vs. Opponent Style Vector.
   Model: XGBoost / Gradient Boosting Classifier.
   Output: Win Probability for specific Tactical Configurations.
4. **Application Layer:** Streamlit Interface for coach interaction and scenario simulation.

Testing:

- Unit testing for feature computation
- Back-testing on historical matches
- Evaluation on unseen teams

7. Deliverables

- **Deliverable 1:** Tactical recommendation system prototype.
- **Deliverable 2**: Simulation and probability estimation engine.
- **Deliverable 3:** Source code repository with documentation.
- **Deliverable 4:** Evaluation results and visualizations.
- **Deliverable 5:** Final report and presentation slides.

8. Timeline / Project Plan

Table: Project Schedule

Phase	Major Tasks	Start Date	End Date
Phase 1: Definition	Problem definition, literature review	1/2/2026	20/2/2026
Phase 2: Requirements	Requirements & dataset preparation	20/2/2026	10/3/2026
Phase 3: Design	Model design & formulation	10/3/2026	15/3/2026
Phase 4: Implementation	Coding, models testing	15/3/2026	5/4/2026
Phase 5: Testing	Integration testing, performance	5/4/2026	15/4/2026
Phase 6: Finalization	Documentation, presentation prep	15/4/2026	1/5/2026

G. Resources and Budget

- **Primary Dataset:** Aggregated historical match data from the "Big 5" European Leagues (Premier League, La Liga, Bundesliga, Serie A, Ligue 1) covering the 2017–2023 seasons. This dataset provides approximately 10,000 matches, ensuring sufficient volume to train a team-agnostic model.
- **Features:** The dataset includes key tactical indicators such as defensive pressure counts, starting formations, expected goals (xG), and possession phases, allowing for the construction of granular tactical profiles.

10. Potential Risks and Mitigation

Potential Risk	Likelihood	Impact	Mitigation Strategy
Tactical recommendations are noisy	Medium	Medium	Feature validation and ablation
Limited causal interpretation	Medium	Low	Clear framing as decision support
Data sparsity for new teams	Medium	Medium	Rolling window aggregation

11. References

[1] Baattite, A. (2023). *Machine learning-based football tactic and style analysis* (Capstone design final report).
[2] Moya, D., Tipantuña, C., Villa, G., Calderón-Hinojosa, X., Rivadeneira, B., & Álvarez, R. (2025). Machine Learning Applied to Professional Football: performance improvement and results prediction. *Machine Learning and Knowledge Extraction*, 7(3), 85.
[3] Joseph, A., Fenton, N., & Neil, M. (2006). Predicting football results using Bayesian nets and other machine learning techniques. *Knowledge-Based Systems*, 19(7), 544–553.
[4] J. Lasek. (2016). Euro 2016 Predictions Using Team Rating Systems. *ECML/PKDD*, pages 21